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Batch :- 1	Class :- SY.BSC.IT
Date of assignment :- 11/08/2026	Date of submission:-11/08/2026

PRACTICAL 14

Aim:- Perform basic time series analysis in scala. Generate synthetic time series data (e.g, daily sales over a month)

Input:-

```
bt  Mag.scala x
import scala.util.Random

object Mag extends App {

    println("Day\tSales")

    for (day <- 1 to 30) {
        val sales = 50 + day * 2 + Random.nextInt(10) - 5
        println(s"$day\t$sales")
    }
}
```

Output:-

```
C:\Users\IT
Day Sales
1 47
2 53
3 51
4 60
5 61
6 59
7 62
8 70
9 66
10 66
11 67
12 69
13 73
14 75
15 78
16 81
17 82
18 81
19 87
20 87
21 93
22 94
```

```
22 94
23 97
24 98
25 100
26 105
27 105
28 110
29 103
30 109
```

Process finished with exit code 0

PRACTICAL 15

Aim:- Create polynomial features from a dataset. Given a list of numbers (e.g, [1,2,3]), Generate polynomial features up to degree 3.

Input:-

```
object Mag extends App {
  val data = List(2, 3, 4, 5, 6)

  for (x <- data) {
    val p1 = math.pow(x, 1)
    val p2 = math.pow(x, 2)
    val p3 = math.pow(x, 3)

    println(s"$x\t$p1, \t$p2, \t$p3")
  }
}
```

Output:-

```
C:\Users\IT-22\.jdk\openjdk-26.0.1\bin\java
2 2.0, 4.0, 8.0
3 3.0, 9.0, 27.0
4 4.0, 16.0, 64.0
5 5.0, 25.0, 125.0
6 6.0, 36.0, 216.0

Process finished with exit code 0
```

PRACTICAL 16

Aim:- Create a scatter plot of random data using Breeze-viz. Label the axes and customize the color of points.

Input:-

```
build.sbt x  Mag.scala
1  libraryDependencies += "org.scalanlp" %% "breeze" % "2.1.0"
2  libraryDependencies += "org.scalanlp" %% "breeze-viz" % "2.1.0"
3
import breeze.plot._
import scala.util.Random

object Mag extends App {
  val x = List.fill(10)(Random.nextDouble() * 10)
  val y = List.fill(10)(Random.nextDouble() * 20)

  println(x)
  println(y)

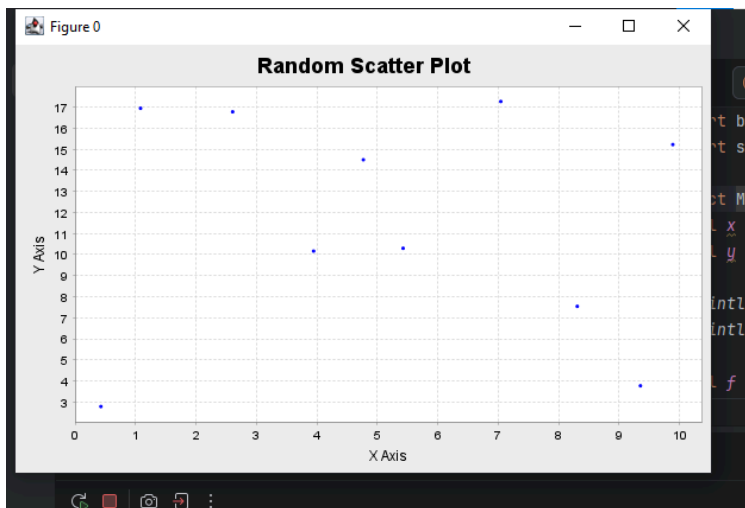
  val f = Figure()
  val p = f.subplot(0)

  p += plot(x, y, '.', colorcode = "blue")

  p.xlabel = "X Axis"
  p.ylabel = "Y Axis"
  p.title = "Random Scatter Plot"

  f.refresh()
}
```

Output:-



PRACTICAL 17

Aim:- Generate a histogram of a dataset using Breeze-viz. Experiment with different bin sizes.

Input:-

```
import breeze.plot._
import scala.io.Source

object d14 extends App {
  val file = Source.fromResource("iris.csv")

  val data = file.getLines()
    .drop(1)
    .map(line => line.split(",")(0).toDouble)
    .toList

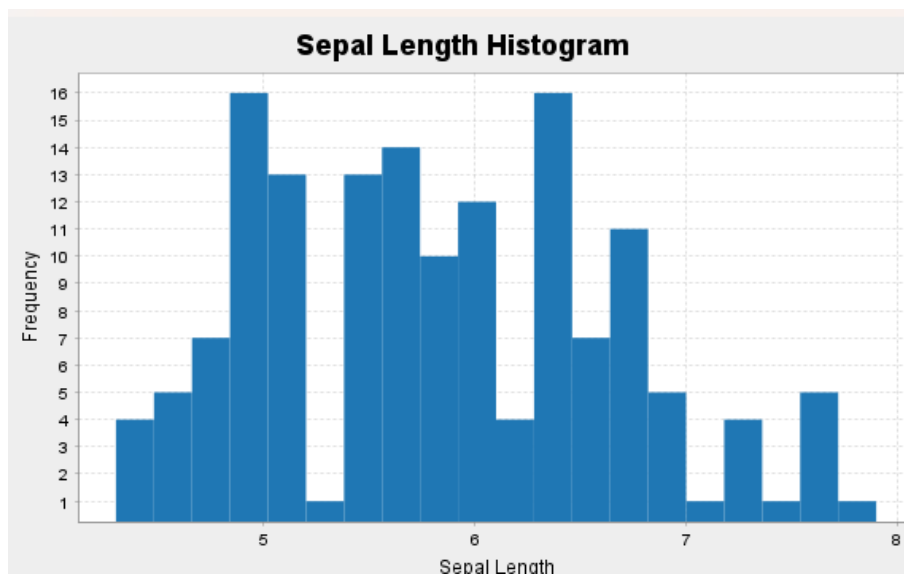
  file.close()

  val fig = Figure()
  val plt = fig.subplot(0)

  plt += hist(data, bins = 20)

  plt.title = "Sepal Length Histogram"
  plt.xlabel = "Sepal Length"
  plt.ylabel = "Frequency"
  fig.refresh()
}
```

Output:-



PRACTICAL 18

Aim:- Plot a line graph for a dataset showing a trend over time.

Input:-

```
import breeze.plot._
import scala.io.Source

> object Demo extends App {

    val file = Source.fromResource("iris.csv")

    val x = file.getLines()
        .drop(1)
        .map(line => line.split(",")(0).toDouble)
        .toList

    file.close()

    val y = x.indices.map(_._2.toDouble).toList

    val fig = Figure()
    val p = fig.subplot(0)

    p += plot(y, x)

    p.xlabel = "Record Number"
    p.ylabel = "Sepal Length"
    p.title = "Sepal Length Trend"

    fig.refresh()
}
```

Output:-

